

Simple Measurement of Magnetic Susceptibility with a Small Permanent Magnet and a Top-Loading Electronic Balance

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Magnetic susceptibility is an important characteristic of materials intimately related to their electronic structure. This is especially true for the salts and complexes of transition metals, because the magnetic susceptibility leads to the knowledge of the magnetic moments of their cations and the number of unpaired d electrons (1, 2). To show how magnetic susceptibility can be measured, a simple, inexpensive, and easy-to-handle device was constructed with a small permanent magnet and a common top-loading electronic balance. It yields results that are accurate enough for educational experiments in high school and freshman courses (3).

Construction of the Device

Figure 1 shows a sketch of the device. Two polyacrylate boxes P_1 and P_2 cover the whole device, protecting it from dust and wind. The magnet M (Fig. 2) is composed of a pair of small neodymium magnets ("Neomax" supplied by Sumitomo Special Metals Co. Ltd., Tokyo;¹ each $20 \times 50 \times 5 \text{ mm}^3$) attached to a sheet of soft iron S ($60 \times 80 \times 5 \text{ mm}^3$) as shown in the figure. A pair of polyacrylate rails R are cemented onto P_2 , so that S can slide along them, carrying M from outside of P to the measuring position in it. A larger sheet of soft iron L ($130 \times 220 \times 5 \text{ mm}^3$) is placed between M and the top-loading electronic balance B (Shimadzu EB-430H; capacity: 430 g, readability: 1 mg), to shield the balance from the effect of the magnetic field of M .

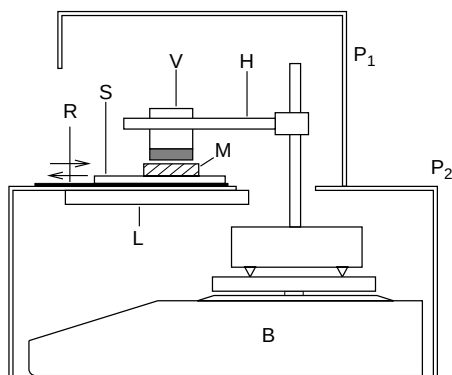


Figure 1. A sketch of the device. P_1 and P_2 : polyacrylate boxes; M : magnet; B : electronic balance; H : sample holder; V : vial; S and L : small and large soft iron plate; R : polyacrylate rails.

The sample holder H is composed of glass rods and polyacrylate plates. It is placed on the pan of B . Similar to a funnel stand, it has an arm with a vertical hole, through which a glass vial V to hold the sample is inserted. The height of the arm (ca. 70 mm from the pan of B) is adjusted so that the bottom of V lies ca. 0.5 mm above the surface of M , when M is at the measuring position. At the same time, the center line of the bottom of V is adjusted to lie above the border of the two small magnets, where the inhomogeneity of the magnetic field is at its maximum.

In a series of measurements, care must be taken to keep the relative positions of all parts of the device unchanged, so that the results of three independent measurements on the same material agree within $\pm 0.001 \text{ g}$.

Measurement of Susceptibilities of Solid Transition Metal Salts

About 1 g of a pulverized solid sample is placed in V at room temperature ($293 \pm 1 \text{ K}$), and its surface is made flat by tapping. First, the exact weight (W) of the sample is measured, keeping M outside of the box. Then, M is slid to the measuring position, where a strong inhomogeneous magnetic field² is applied to the sample. This brings about a notable increase ΔW in its apparent weight, owing to the paramagnetism of the transition metal ions; for example, the values of ΔW for 1.012 g of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ and 0.969 g of $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ were

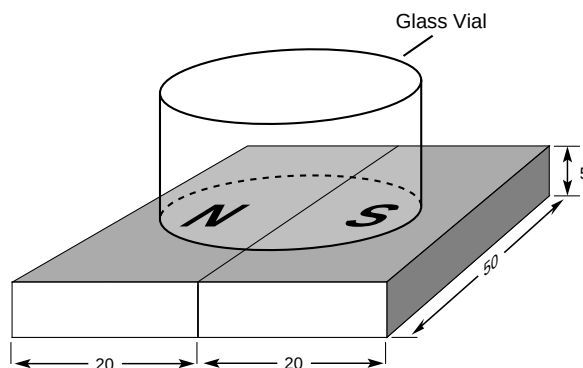


Figure 2. Enlarged view of M . Note that the poles of the two small magnets are on their flat top and bottom faces, not on their ends.

Table 1. Values of χ and μ for Some Representative Transition Metal Salts

Salt	Experimental		Literature (4)		Spin-Only μ/μ_B	<i>n</i>
	$\chi/10^6 \text{ cm}^3 \text{ g}^{-1}$	μ/μ_B	$\chi/10^6 \text{ cm}^3 \text{ g}^{-1}$	μ/μ_B		
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$	7.03	2.0	5.85	2.0	1.73	1
$\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$	9.78	2.0	8.33	2.0	1.73	1
$\text{Cu}(\text{OAc})_2 \cdot \text{H}_2\text{O}$	4.56	1.5 ^a	4.07	1.4 ^a	1.73	1
$\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$	19.7	3.5	15.8	3.2	2.83	2
$\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$	17.8	3.2	16.9	3.1	2.83	2
$\text{CoSO}_4 \cdot 7\text{H}_2\text{O}$	36.0	4.9	33.1	4.9	3.87	3
$\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$	39.7	4.7	40.8	4.9	3.87	3
$\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$	78.5	6.0	73.8	5.9	5.92	5
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	55.7	5.9	56.4	5.7	5.92	5
$\text{K}_4[\text{Fe}(\text{CN})_6] \cdot 3\text{H}_2\text{O}$	-0.48	0	-0.37	0.11	0	0 ^b
$\text{K}_3[\text{Fe}(\text{CN})_6]$	7.40	2.5	6.96	2.26	1.73	1 ^b

^aSubnormal values owing to Cu–Cu interaction (4).^bLow-spin complexes (4).

+0.077 g and +0.566 g, respectively, when they were measured in a vial with diameter 30 mm, bottom thickness 2 mm, and weight 8.613 g.

The magnetic susceptibility of the sample, χ_S , is now calculated from the values of W and ΔW with the following equation, making use of the susceptibility of Mohr's salt, $\chi_R = 31.6 \times 10^6 \text{ cm}^3 \text{ g}^{-1}$, as the reference:

$$\chi_S = \chi_R [W_R / (\Delta W_R - \Delta W_E)] [(\Delta W_S - \Delta W_E) / W_S] \quad (1)$$

Here the subscripts R and S refer to the reference (Mohr's salt) and the sample, respectively, and ΔW_E is the value for the empty vial (in this case, -0.026 g). Table 1 shows the χ values of some representative transition metal salts obtained in this way, and their magnetic moment μ (in Bohr magnetons) calculated with the well-known equation,

$$\mu = 2.83 [\chi_M T]^{1/2} \quad (2)$$

Here χ_M is the molar susceptibility ($\chi \times$ formula weight), corrected slightly for the diamagnetism of the component ions and molecules; T, the temperature in Kelvin.³ Comparison with the published values (4) shows that the χ values obtained in this way are reasonably good, and the μ values are fairly correct, considering the simplicity of the handmade device, possible experimental errors, and approximations involved in the calculations. The comparison of the μ values obtained with the "spin-only" values

$$\mu = [n(n+2)]^{1/2} \quad (3)$$

leads to the determination of the number *n* of unpaired d electrons, as is well described in textbooks (5).

This device can also be applied to diamagnetic materials, where the value of W decreases slightly in a magnetic field, leading to small negative values of ΔW and χ . The latter agree again reasonably with published values (4); for example, the value of ΔW for 0.985 g of water is -0.038 g ($\chi/10^6 \text{ cm}^3 \text{ g}^{-1}$: exptl, -0.83; lit., -0.72), and that for 0.989 g of NaCl is -0.036 g ($\chi/10^6 \text{ cm}^3 \text{ g}^{-1}$: exptl, -0.53; lit., -0.52). The value of ΔW_E (-0.026 g) for the vial indicates that glass is a diamagnetic substance, too.

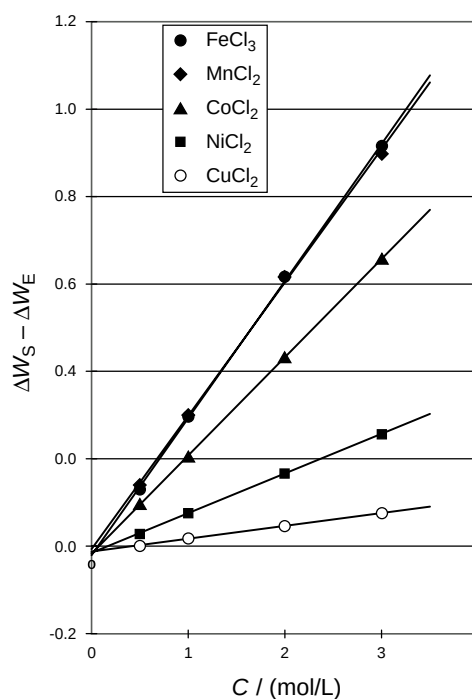


Figure 3. Linear relationship between $(\Delta W_S - \Delta W_E)$ and C for CuCl_2 , NiCl_2 , CoCl_2 , MnCl_2 , and FeCl_3 solutions. The gradients α for them are 0.314, 0.305, 0.225, 0.091, and 0.029, respectively.

Measurement of Susceptibilities of Solutions

Magnetic susceptibilities of solutions ($\geq 1 \text{ mol dm}^{-3}$) of transition metal salts or complexes can also be measured with this device. One milliliter of the solution is pipetted into V, and the value of its ΔW_S is measured as in the case of the solid. When $\Delta W_S - \Delta W_E$ is plotted against concentration C , the results in Figure 3 are obtained for CuCl_2 , NiCl_2 , CoCl_2 , MnCl_2 , and FeCl_3 solutions. In every case the relation is expressed as a straight line which, starting from ca. -0.012 g at $C = 0$ (the value for pure water), increases steadily with a gradient α . This α , which increases from CuCl_2 to FeCl_3 with the increase of *n* in their cations, can now be taken as a measure of the χ of the solution, and its square root as a measure of μ . The latter relationship can be clearly seen from the fact that the values of $10.57(\text{gradient})^{1/2}$ for these solutions are 5.92, 5.84, 5.01, 3.18, and 1.81, respectively, agreeing quite well with the μ values obtained with solid salts (Table 1). So it is possible to estimate the μ and *n* of any other salt or complex in solution, if one can plot its $(\Delta W_S - \Delta W_E)$ against C and obtain its gradient α .

Further Applications

This device can be applied further to observe the magnetochemical changes in solution. For example, it shows that the paramagnetism of an aqueous solution of Fe^{2+} disappears when an ethanolic solution containing 3 moles of 1,10-phenanthroline is added to it. A similar change is found in an aqueous ammoniacal solution of Ni^{2+} upon the addition of 2 moles of dimethylglyoxime in ethanol. Deep red

complexes $[\text{Fe}(\text{phen})_3]^{2+}$ and $[\text{Ni}(\text{dmg})_2]^{2+}$, which are diamagnetic and "low spin", are formed in these changes (5). The former remains in solution, but the latter separates out immediately, so the device is applicable even to such a heterogeneous system.

All these results prove the reliability and usefulness of this device as a tool for introductory demonstrations and student experiments in magnetochemistry. If only qualitative and semiquantitative results are required (for example, the difference between diamagnetic and paramagnetic substances, and general increase of the magnetic moment with increasing n), the operations can be simplified accordingly.

An additional merit of this device is that, when not in use, it can be easily disassembled and the top-loading balance can be used as such.

It may be added that, although many papers on magnetochemical experiments are found in this *Journal* (6), none of them describe an apparatus similar to the one in this paper.

Acknowledgments

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Notes

1. U.S. Office: 23326 Hawthorne Blvd., Suite 360, Skypark 10, Torrance, CA 90505; Tel. 310/378-7886; Fax 310/378-0108.
2. The values of maximum flux density and magnetic field gradient, measured near the bottom of V with a Shimadzu GK-3 magnetic fluxmeter, were ≥ 0.34 T and ≥ 0.7 T cm^{-1} , respectively (3).
3. The values of such corrections (χ_M , in 10^6 $\text{cm}^3 \text{mol}^{-1}$) used to obtain the $\chi_M(\text{exptl})$ in Table 1 are Na^+ , -5; K^+ , NH_4^+ , and Fe^{2+} , -13; Fe^{3+} , -10; Mn^{2+} , -14; Co^{2+} , -12; Ni^{2+} and Cu^{2+} , -11; Cl^- , -26; SO_4^{2-} , -40; CN^- , -18; H_2O , -13 (1).

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5. See standard textbooks of inorganic chemistry; e. g., Cotton, F. A.; Wilkinson, G.; Gaus, P. L. *Basic Inorganic Chemistry*, 3rd ed.; Wiley: New York, 1995; p 2. Jolly, W. L. *Modern Inorganic Chemistry*, 2nd ed.; McGraw-Hill: New York, 1991.
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